Container Diagram - DocuFlow: A Secure Document Tracker with Multi-Level Review System for DOST-STII

DOST-STII
[enterprise]

«person»
Document Source
DOST-STII staff member who submits documents, tracks their status, and resubmits returned documents or cancels a submission to correct it.

«person»
Immediate Supervisor (L1)
Reviews documents at Level 1 (returns or forwards them) and can also submit documents.

«person»
Section Head (L2)
Reviews documents at Level 2 (returns or endorses them) and can also submit documents.

«person»
Division Chief (L3)
Gives the final review (returns or approves documents) and monitors the system-wide summary.

«person»
Super Admin
Checks the audit trail and manages user accounts and roles, the holiday calendar, and AI knowledge and settings.

Manage Account Access
Manage Documents (Submit/Resubmit Document)
[Views in-system notifications]
[HTTPS]

Manage Account Access
Manage Documents (Submit/Resubmit Document, Review Document)
[Views in-system notifications]
[HTTPS]

Manage Account Access
Manage Documents (Submit/Resubmit Document, Review Document)
[Views in-system notifications]
[HTTPS]

Manage Account Access
Manage Documents (Review Document)
[Views in-system notifications]
[HTTPS]

Manage Account Access
Manage System Administration (audit trail, user accounts and roles, holiday calendar, AI knowledge and settings)
[HTTPS]

DocuFlow
[system]

«container»
Web Application
[React]
Runs in the browser. Shows only the sections each role can use: My Documents, Review Queue, All Documents, and Submit Document, plus the search bar, the review screen with the AI feedback, the profile, and in-system notifications. For the Super Admin: Audit Trail, User Accounts, Holiday Calendar, and AI Settings.

Sends requests and receives system data
[HTTPS / JSON]

«container»
Backend Application
[Laravel / PHP (Apache application server)]
Handles Sign in with Google and role-based access, submissions, resubmissions, and cancellations, reference numbers, duplicate checks by hashing, review routing (L1, L2, L3), moving documents when a reviewer changes, TAT in working days, file downloads, AI knowledge review, PDF reports and audit exports, notifications, and audit logging. Runs AI requests and emails as background jobs, and runs a daily scheduled check for overdue reviews.

«container»
Uploaded File Storage
[Private Laravel file storage (application server disk)]
Keeps uploaded PDF, DOCX, and XLSX files (up to 10MB each and 100MB per submission) exactly as submitted. Files are opened or downloaded only through the backend after an access check.

Saves and retrieves uploaded files
[Laravel Filesystem]

Reads and writes system records
[SQL]

«container»
DocuFlow Database
[MySQL (database server)]
Stores user accounts and roles with the encrypted Google Drive permission, documents and their revisions, Google Workspace links, file details and locations, document hashes, review records and remarks, AI feedback, AI knowledge and settings, the holiday calendar, in-system notifications, the audit trail, and the background job queue.

Redirects to Sign in with Google; opens Google Workspace links in a new tab
[HTTPS]

Sign in with Google, link sharing checks, reading linked documents, and email through Gmail
[OAuth 2.0 / Google Drive API, SMTP]

Sends document text and AI knowledge, receives feedback per review level
[HTTPS / REST API]

«external_system»
Google Workspace
Handles Sign in with Google, checks the sharing of Google Workspace document links, and delivers email notifications through Gmail.

«external_system»
AI Assistant
AI model accessed through an API that gives preliminary feedback and suggestions for each review level.